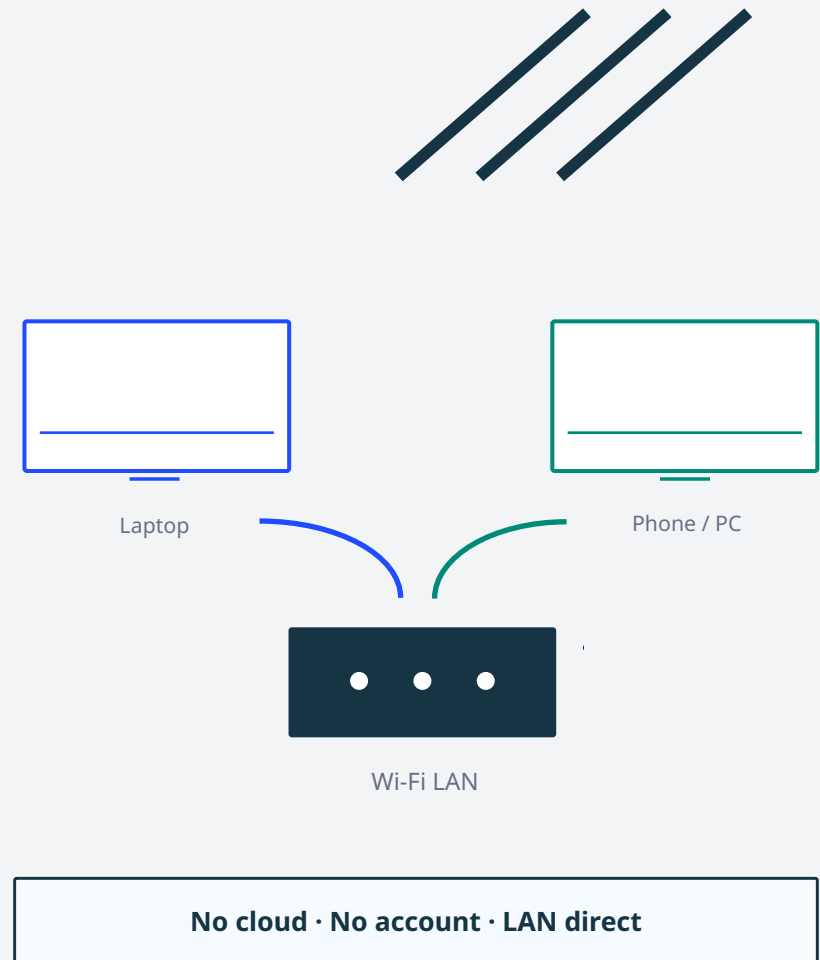


SafeDrop

Secure-Zero-Config LAN file & Clipboard Sharing Tool

CNL - Group 5 Final Project Proposal

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Motivation - 跨裝置傳輸

生態系綁架、便利性

手機、平板、Windows/Mac 之間常需傳文字片段、照片、連結或程式碼等等各式檔案

效率與隱私

Line 等社群軟體 / Email / 雲端硬碟需要連線至外部網際網路網、需要帳號、且檔案需要經過第三方平台

操作繁瑣

USB 或共享資料夾不直覺，自行架設 Socket Server 需要手動查詢並輸入 IP 對使用者不友善

□ 同一個 Wi-Fi 內，其實可以直接透過區域網路高速傳輸！

Cloud

帳號/外網

USB

線材

Messaging

第三方平台

SafeDrop

區網直傳

Goal -- *make nearby sharing feel like one click*

Problem Definition

同一 Wi-Fi 下，如何做到「不用設定」且「安全」分享？

核心問題

如何在無需帳戶、雲端服務或手動 IP 配置的情況下，讓附近的設備安全地共享資料？

- 自動發現附近裝置
- 直接傳送檔案、文字、連結、程式碼片段等
- 不依賴外部雲端 Server
- Receiver 端必須能接受或拒絕
- 可配對與加密，避免同網段竊聽

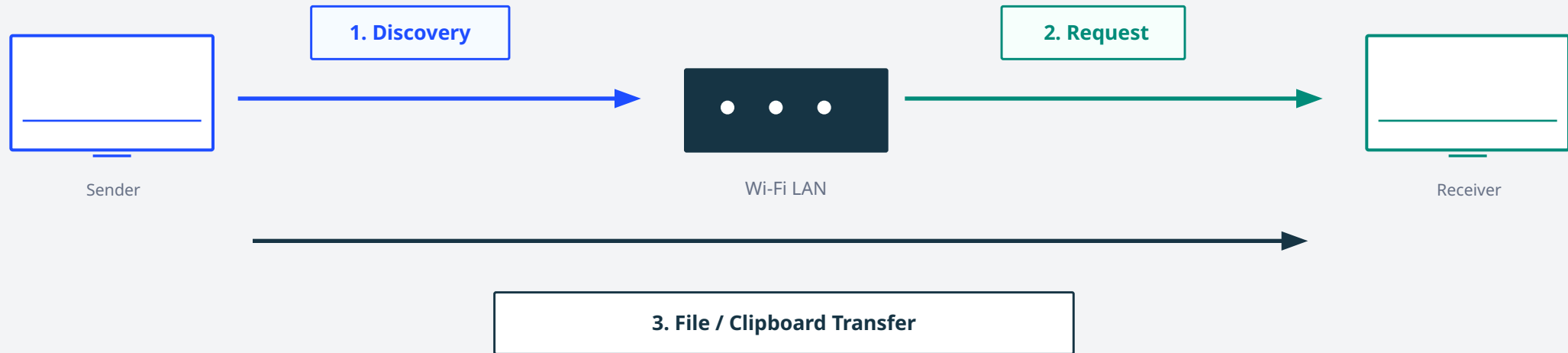
Zero-configuration

LAN direct

Privacy-friendly

Proposed Solution: SafeDrop

類似 AirDrop / Nearby Share, 但面向一般區網環境與跨平台原型



SafeDrop lets devices in the same LAN discover each other automatically and exchange data through a direct TCP connection.

Key Features

奠定基礎通訊架構，強化安全與使用者體驗



01

Zero-Configuration Discovery

不需手動輸入IP；同一 Wi-Fi 裝置自動出現

02

Direct LAN File Transfer

使用 TCP Socket 穩定傳送，顯示進度與速度

03

Clipboard Sharing

文字、URL、code snippet 一鍵傳到另一台裝置

04

Receiver Confirmation

每次傳輸都需要接收端接受或拒絕

05

Security Extension

配對碼、信任裝置、加密傳輸

System Architecture

從 UI 到網路傳輸的分層設計

User Interface

裝置列表、傳送面板、接收確認、進度顯示

Device Discovery

UDP Broadcast 探索區網內服務

Control Protocol

JSON messages: metadata、request、clipboard content

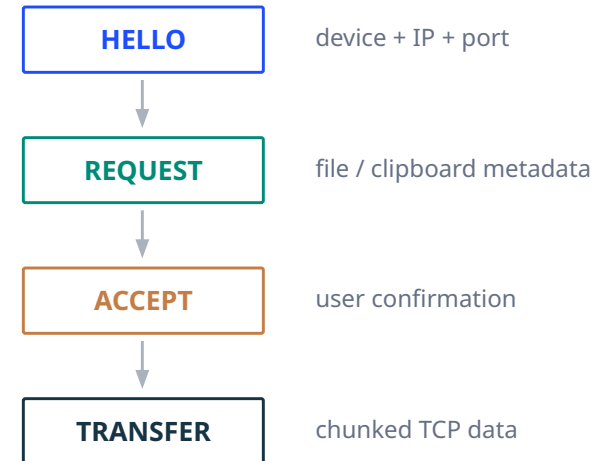
Data Transfer

TCP Socket: Reliably 傳送檔案與文字內容

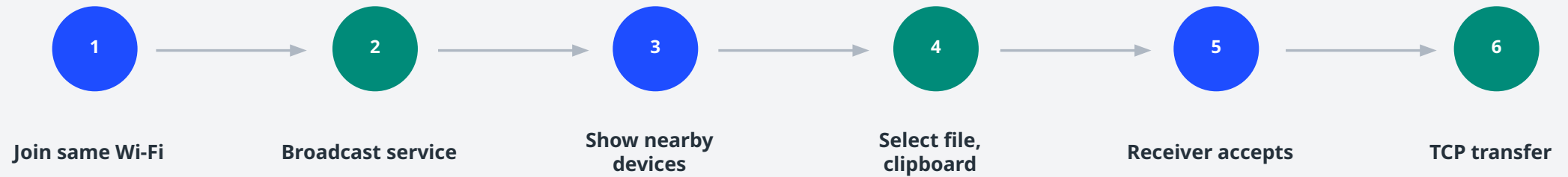
Security Layer

Pairing code、trusted devices、encryption

Protocol overview



Workflow



Demo focus: Two laptops in the same Wi-Fi network discover each other, send a PDF, then send a URL/code snippet through clipboard sharing.

Clipboard Sharing Design



Sender preview

`https://github.com/example/project`
`# code snippet or useful text`

Preview before sending

Send to selected
device

JSON over TCP



Receiver confirmation

Alice wants to send clipboard content.
Type: URL / Text / Code

Accept?

Copy to clipboard

Open URL

Technical Challenges & Solutions



Challenge

Solution

UDP broadcast may be blocked



Fallback: manual IP, QR code pairing, or mDNS

Shared Wi-Fi security risk



Receiver confirmation, pairing code, trusted devices, encryption

Large file interruption



Chunk-based transfer and possible resume support

Clipboard privacy



No background auto-sync; preview content before sending



Demo Plan & Development Schedule



Demo Plan

- 兩台筆電連到同一 Wi-Fi, 自動發現對方
- 傳送 PDF / 圖片, 顯示進度與速度
- 傳送 URL 或程式碼片段, 接收端一鍵複製
- 展示接受/拒絕流程; 可選用 Wireshark 觀察加密效果

Timeline

W1

Discovery + TCP prototype

W2

GUI + clipboard sharing

W3

Pairing + encryption

W4

Testing + final demo

Expected outcome : a working LAN sharing system demonstrating UDP discovery, TCP transfer, user confirmation, and secure local communication.